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Industrial big data-driven mechanical performance prediction for hot-rolling steel using lower upper bound estimation method

Gongzhuang Peng [a,*], Yinliang Cheng [a], Yufei Zhang [b], Jian Shao [a], Hongwei Wang [b], Weiming Shen [c]

[a] National Engineering Research Center for Advanced Rolling Technology, University of Science and Technology Beijing, Beijing 100083, China
[b] ZJU-UIUC Institute, Zhejiang University, Haining, China
[c] The State Key Laboratory of Digital Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan 430074, China

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ABSTRACT

Industrial big data technology has become one of the important driving forces to intelligent manufacturing in the steel industry. In this study, the characteristics of data in steel production are analyzed and an industrial big data platform for steeling process is developed to extract the quality-related parameters. A data-driven approach to construct prediction intervals (PIs) of mechanical performances for hot-rolling strips is proposed to represent the uncertainty and reliability of the prediction results. The proposed method employs a new manifold visualization method, SLISEMAP, to reduce the feature dimensions with interpretability, utilizes lower upper bound estimation (LUBE) method to obtain the PIs, in which the broad learning system (BLS) is used as the basic training network model and the artificial bee colony (ABC) algorithm is applied to optimize the weighting parameters of BLS under the LUBE framework. A hot-rolling steel coil dataset consisting of 39 variables and 1335 coil samples is used to validate the proposed method. Two Delta-based approaches, namely back propagation neural network (BPNN) and extreme learning machine (ELM); and three LUBE-based approaches, namely ABC-BPNN, ABC-ELM, and ABC-support vector regression (SVR) are compared with the proposed method. Results show that the proposed ABC-BLS in LUBE is effective and efficient in constructing the PIs with a higher coverage probability and a narrower interval width.

1. Introduction

With the maturity of technologies such as advanced sensing, wireless communication, and the internet of things, the integration of information technology and manufacturing systems is accelerating and huge amounts of industrial data have been generated and collected over life cycles of products. Industrial big data has become one of the important driving forces of intelligent manufacturing [1,2]. Industrial big data analytics helps enterprises obtain timely information regarding the status systems and promote optimal decision-making, which enhances manufacturing competitiveness. Industrial data collected from the manufacturing system covers market requirements, product development, manufacturing and assembly, equipment operation, quality inspection, planning and logistics, and after-sales service [3]. Big data analysis technology helps to mine potential knowledge from structured and unstructured data in the above process, and has been applied widely in many fields, such as process optimization [4], process monitoring [5], equipment maintenance [6], and energy consumption management [7].

The manufacturing industry can be divided into two types: discrete and process industry, of which the steel industry is a typical representative of the process industry. Under the rising resources and environmental constraints, the newly proposed emission peak and carbon neutrality targets require the steel industry to achieve higher-quality development with the help of next-generation information technologies such as artificial intelligence and big data [8]. Big data technology plays a vital role in the development of intelligent manufacturing in the steel industry. On the one hand, it is difficult to accurately describe the complex physical and chemical reaction processes in steel production using mechanism models, and some process parameters cannot be measured online; on the other hand, modern steel enterprises have established complete information systems and accumulated a large amount of historical data that has not been effectively used [9]. Mining tacit knowledge from big data in the steel industry can realize the optimization of comprehensive production indicators such as product

* Corresponding author.
E-mail address: gzpeng@ustb.edu.cn (G. Peng).

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quality, output, cost, and consumption in the entire production process. Prediction of the quality of steel products such as mechanical properties has received great attention from both the academia and industry [10]. An accurate prediction model realizes rapid reverse optimization of process parameters and reduces the number of inspections by establishing the mapping relationship between influencing factors and quality indicators, thereby speeding up the development of new products and reducing research costs.

Existing data-driven quality prediction models in steel production are mainly based on point estimation. However, lots of uncertain factors in industrial data bring uncertainty to forecast results. Furthermore, engineers need reliability of prediction results to make decisions on product quality and process optimization. Data uncertainty is usually modeled by probability distributions or prediction intervals. In this study, the uncertainty of the forecast results is characterized by means of the prediction interval to obtain the numerical estimate and its reliability. Thus, it is necessary to study the estimation of prediction intervals (PIs) related to mechanical properties. Lower upper bound estimation (LUBE) is one of the most popular PIs construction methods, which makes no assumptions about the distribution of predicted targets and does not calculate complex intermediate matrices. The core of the LUBE method are the network model and the optimization algorithm. In this study, we construct the PIs for mechanical performance based on a newly developed fast and efficient network called broad learning system (BLS) [11–13] and apply the artificial bee colony (ABC) algorithm to optimize the weighting and structural parameters of BLS under the LUBE framework.

The remainder of this paper is organized as follows. Section 2 introduces the industrial big data platform for steel production. The data source and feature extraction methods are described in Section 3. The formulation of BLS and LUBE methods, as well as the mechanical performance prediction model based on the ABC-BLS are explained in detail in Section 4. Computational experiments are described in Section 5, through which comparative studies were conducted. Finally, the concluding remarks are presented in Section 6, along with a discussion on potential areas of future research.

2. Literature review

The traditional mechanical performance prediction of hot-rolling strips is estimated by empirical formula through mechanism models combined with experimental analysis. However, the complex and nonlinear production process brings difficulties to the accurate establishment of the mechanism models. In recent years, industrial big data technology has gradually emerged in the steel industry. Cao et al. [14] developed a blockchain-based steel IoT quality traceability system with the use of the alliance chain mode and the Hyperledger blockchain platform, which effectively avoid the incomplete information and trace the quality of steel products. Zhang et al. [15] proposed an IoT system architecture for the continuous steel casting production line, which addressed the challenge of heterogeneous data and multiple communication protocols.

Data-driven mechanical performance prediction of steel products has been studied in recent years [16–19]. Si et al. applied the Pearson correlation to select key features and random forest regression model to establish the relationship between features and material properties of Zr-Ti-Nb-O alloys, including compression strength and fracture strain [20]. Kim et al. [21] employed the variational autoencoder to reconstruct the microstructure images and Gaussian process regression to predict the mechanical properties of dual-phase steel based on microstructural characterizations such as volume fraction, morphology, and spatial distribution. Xie et al. [22] utilized an optimal deep neural network to predict the mechanical properties of steel plates online based on the process parameters and material compositions. However, most of the existing data-driven performance prediction models are point prediction methods and convey little information about the uncertainty of

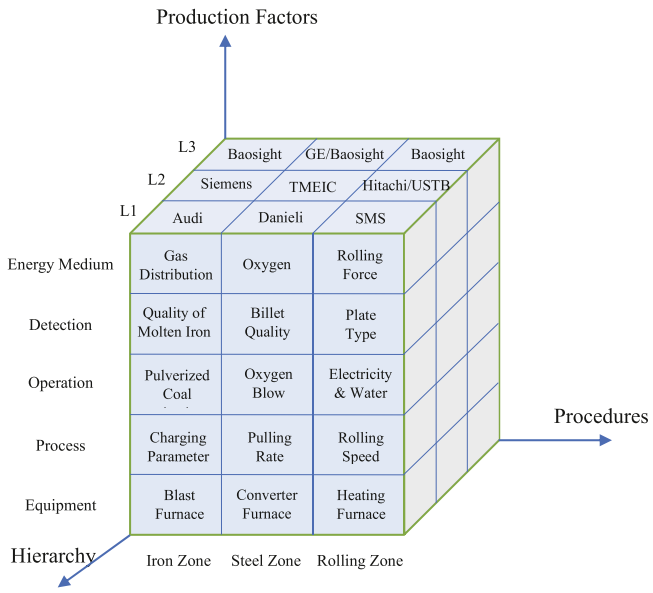


Fig. 1. Multi-sources of big data in steel industry.

prediction errors and the reliability of prediction results. In addition, the user requirements for mechanical properties of steel products are generally in the form of intervals. For example, an industry requirement for the yield strength of marine steel is 550–690 MPa. Therefore, to obtain prediction results with a certain degree of confidence and deal with the uncertainty in industrial production, it is of great significance to study the construction of PIs.

PIs construction is a current hot spot in the domain of prognostic and health management [23] and energy forecast [24]. The current PIs construction methods fall into the following five categories: mean variance estimation (MVE) [25], delta [26], bootstrap [27], Bayesian [28], and LUBE [29]. The MVE method has a relatively low coverage due to the underestimation of the data variance. The delta method constructs the PIs through a Taylor series expansion, which requires the calculation of the Jacobian and Hessian matrix, which is computationally intensive. Bayesian method infers the future state of data through statistical probability information of historical data, and its accuracy largely depends on prior knowledge. The LUBE method predicts the upper and lower bounds of unknown targets by generating a neural network model with two outputs. It makes no assumptions about the data distribution and avoids the problem of numerical calculation of Jacobian and Hessian matrices. Since the LUBE method cannot construct a supervised learning model, the structure and parameters of the network must be tuned with a parameter optimization algorithm. Yan et al. [30] proposed a prediction model of LUBE based on a deep residual neural network, which predicts flood flow in intervals according to historical and future rainfall and historical flow. Zhou et al. [31] constructed a LUBE interval prediction model based on the improved long short-term memory neural network to predict wind power, which processes time series characteristics of wind power data. Kabir et al. [32] proposed a customizable smooth cost function considering heteroskedasticity, asymmetry and fault distance to construct a LUBE forecasting model and applied it to power demand, wind power generation and temperature forecasting, respectively.

To the best of the authors' knowledge, few studies have applied the LUBE method to construct PIs for steel mechanical performances. The difficulty of LUBE model is to design an efficient neural network to perform point prediction according to the characteristics of the dataset, and to optimize the parameters of the network according to the evaluation indicators of PIs, which include PI coverage probability and interval widths.

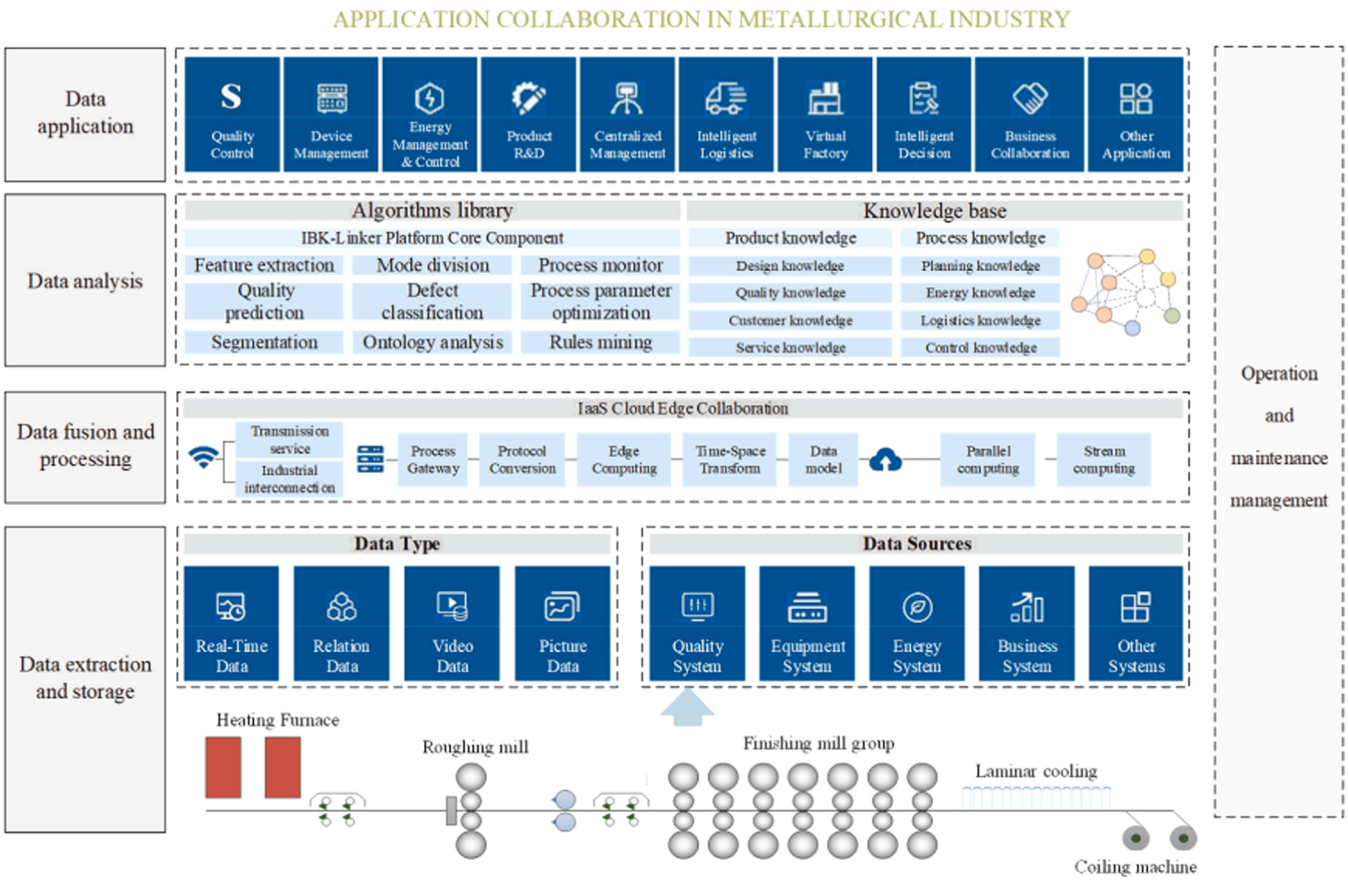


Fig. 2. Framework of the big data platform in steel industry.

3. Industrial big data platform for steeling process

3.1. Characteristics of big data in steel industry

The iron and steel production process has the features of multi-process continuous production, strong inter-process heritability, and nonlinearity of various influencing factors. The related steel big data has the following typical characteristics:

1) Multi-sources: As shown in Fig. 1, the data in steel industry mainly comes from the five-layer information architecture, including basic automation system, process control system, manufacturing execution system, enterprise resource planning system, and decision-making support system. From the perspective of sub-processes, the data sources include the iron-making process, the steelmaking process, the hot-rolling process and the cold-rolling process. From the perspective of business fields, the data covers equipment management, production process, quality control, product sales, logistics supply, and finance. The problem of incompatibility between information systems has caused serious "information islands", and the heavily-fragmented data impedes exploiting business value.
2) Heterogeneity: The steel production process generates a variety of data in different structures and types. The industrial big data in steeling process include the following types: numerical structured data, such as set points and chemical composition of process parameters; image data, such as burden surface, slab surface cracks, and strip surface defects; sound data, such as rolling mill vibration and manual commands; time series data, such as temperature fluctuation curves and slab offset curves; text data, such as user requirements and process operation logs. Heterogeneity complicates the management and use of industrial big data.
3) Spatial-temporal hereditary: The process relativity and quality have hereditary characteristics between different processes. For example, parameters such as the composition and temperature of the molten iron in the ironmaking process affect the smelting process, cost, and control method of the downstream steelmaking process. The shape and surface quality of the hot rolling process have a direct effect on the shape and surface of the cold rolling process. In terms of production planning, the scheduling of downstream rolling plans is determined by steelmaking plans and logistics scheduling.
4) Multi-mode and time-varying: Steel production is a complex process with traits of multi-variable coupling and nonlinear time-varying. Due to the fluctuation of raw material composition and equipment status and the need for customization, steel production also has variable working conditions and multi-mode characteristics. Different working conditions have their own production process and parameter requirements.

The above features complicate the application of big data in the steel industry in terms of data acquisition, data storage and data analysis.

3.2. Architecture of big data platform in steel industry

Fig. 2 illustrates the four-layer architecture of big data platform in steel industry.

3.2.1. Data extraction and storage layer

The sources of steeling big data include PLC, DCS, MES, ERP; other multi-level control systems for sub-processes such as continuous casting, hot rolling, cold rolling, and acid rolling; and also supporting equipment such as grinding rolls and instruments. Since the control system comes from different integrated manufacturers, the communication protocols,



Fig. 3. Digital twin model of the hot-rolling line.

network architectures, and hardware interfaces are diverse. In the developed big data platform in steel industry, a data acquisition system compatible with multiple protocols and interfaces is constructed. PLC control system data is collected through the standard OPC protocol provided by a real-time database; relational data and pictural data in L2, MES, ERP systems and meter instruments are collected based on the ODBC interface; the equipment status, spare parts and raw materials that are not synchronized in the MES and ERP system are accessed through a custom human-computer interaction interface.

In terms of data storage, real-time databases (e.g. PI, Wonderware) are used to store high-frequency rolling process data, relational databases (e.g. Oracle, SQL server) to store structured data and model parameters, time series databases (e.g. InfluxDB) to store batch-related multi-dimensional time-series data, non-relational databases (e.g. MongoDB) to store unstructured data such as defect pictures and vibration audio, in-memory databases to store data with high processing efficiency requirements, and data files to store offline batch data.

3.2.2. *Data fusion and processing layer*

Due to the heritability of steel production, it is necessary to perform spatial-temporal transformation of the data on the edge side. Steeling data is aligned and converted between multiple processes based on metallurgical mechanism and process characteristics to achieve synchronization of quality and process parameters along the strip length. On the one hand, the mapping of the material spectrum between furnaces, slabs, hot-rolled steel coils, and cold-rolled steel coils is realized through space-time matching. On the other hand, considering the inversion of the top and bottom of the material and the removal of the head and tail sides, the data traceability can be realized through spatial-temporal transformation. The data generated in the steeling process is prone to noise, missing values, and data inconsistency due to sensor failures, human operation factors and disordered network transmission. Using these data directly for data analysis severely impacts the accuracy and reliability of the model. Therefore, it is often necessary to preprocess the data to eliminate noise, correct inconsistencies, and identify outliers to improve model robustness and prevent model overfitting.

3.2.3. *Data analysis layer*

Optimization decisions to improve production efficiency and product quality in the steel industry are made through in-depth mining and accurate modeling of big data. According to the characteristics and application requirements of steeling data, the developed platform divides data analysis methods into the following six categories: feature extraction, mode division, process monitoring, quality prediction, defect classification, and process parameter optimization.

Feature extraction uses feature selection methods such as the random forest algorithm or dimensionality reduction methods such as principal component analysis to reduce the dimension of process parameters to avoid overfitting problems. Mode division identifies multiple stable operating points in the production process through clustering algorithms and can also merge samples with similar process parameters or mechanical properties. Process monitoring uses relevant statistical indicators to monitor and determine the fault status of the sampled online data. Quality prediction establishes the mapping relationship between measurable quality indicators, material components, and process parameters. The measurable indicators include temperature, material size (thickness, width, etc.), convexity, and mechanical properties. When the quality indicators are not measurable, such as surface defect, pressure oxidation defects and warping, classification algorithms are used to classify the defect types of the predicted indicators. Parameter optimization aims to obtain a reasonable range of process parameters associated with product quality and cost by mining excellent samples through optimization algorithms.

3.2.4. *Data application layer*

Distributed application services are rapidly developed for the needs of different scenarios in steel enterprises, making various applications such as quality, equipment, energy, production, and safety more intelligent and convenient. Through correlation analysis of process parameters and quality indicators in the rolling process, control and process optimization are realized, thereby improving the product quality. Through collection and analysis of sensing and service data in roller shops, remote inspection and intelligent detection of equipment health are realized, and suggestions are provided for hot-rolling scheduling and roller-change decision-making. Through monitoring, statistics and analysis of energy data including electricity, gas, steam, oxygen, etc., balanced distribution and optimal scheduling of energy and byproducts are realized, which contributes to green production. Through analysis and matching of order data and inventory data, the production scheduling plan is optimized, and order response speed and turnover rate of the warehouse are improved.

The platform builds a digital twin model of steel production based on industrial big data, in which multi-dimensional, multi-scale and multi-disciplinary virtual models are constructed to simulate and depict the physical entities in real environment. Through real-time interaction and iterative optimization of virtual models and physical entities, precise prediction and control of the process, self-organized optimization and scheduling of production, management of the entire life cycle of equipment, and traceability and control of product quality are further improved. Fig. 3 shows the digital twin model of the hot-rolling line in

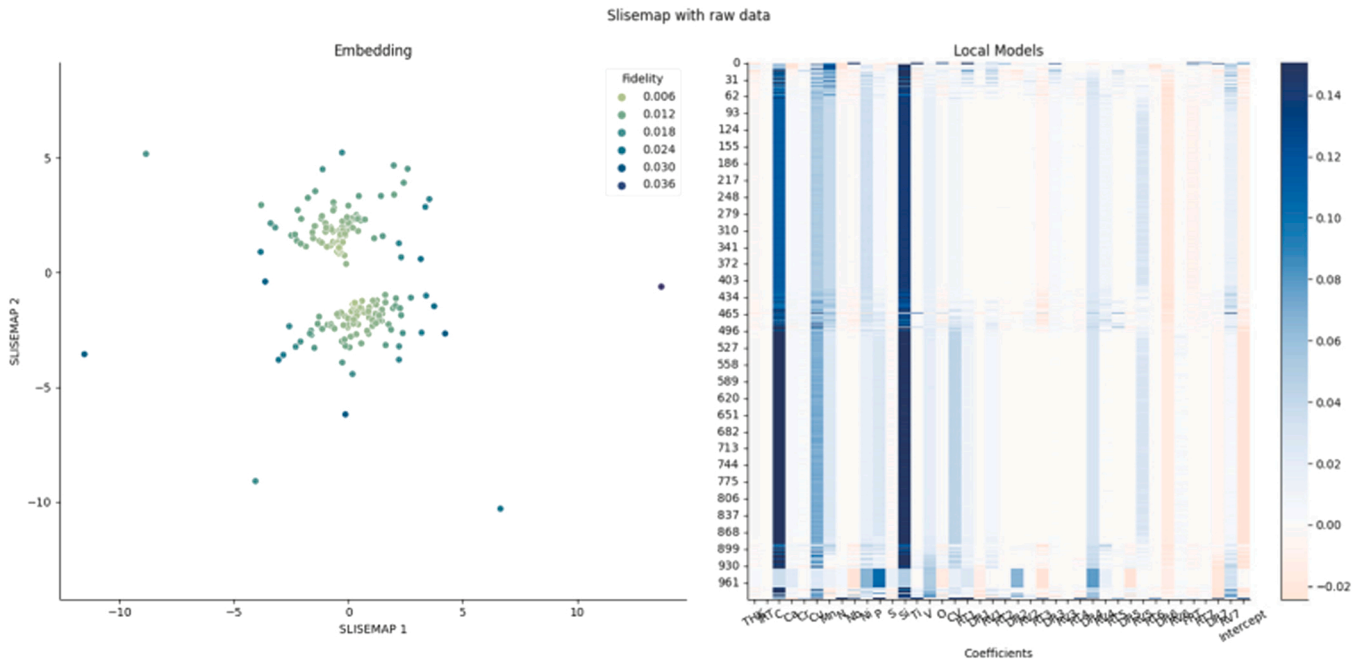


Fig. 4. Dimensionality reduction results by SLISEMAP.

the application layer of the platform.

3.3. Data preprocessing for mechanical performances prediction

The mechanical properties of materials mainly depend on the chemical composition of the material and its internal structure. The macroscopic mechanical properties are the external manifestations of the internal microstructure of the material. Therefore, to accurately predict the mechanical properties of materials, it is necessary to analyze the evolution process of their internal microstructure as well as the parameters that cause these evolutions. According to the Hall-Petch formula, the yield strength of metallic materials increases with decreasing grain size. During the production of the strip, the grains will go through the grain growth stage in the heating furnace, the recrystallization stage in the rolling process, and the phase transformation stage in the cooling process. The heating furnace parameters include heating time and heating temperature, the rolling process parameters include rolling temperature and deformation rate, and the cooling process parameters include cooling rate. The chemical composition that affects the characteristics of the material is greatly related to the change of the grain. Based on the above analysis, we selected 39-dimensional variables as the input of the prediction model. Three representative mechanical properties—strength, tensile strength and elongation—were selected as outputs of the model.

Dimensionality reduction, in which the indefinite features are removed or multiple features are combined to make the data more interpretable, is an important measure in dealing with high-dimensional data. In this study, we adopted a newly proposed explainable dimensionality reduction approach, SLISEMAP [33], which makes the supervised learning model of the data points in the embedding space explained by a combination of interpretable local model. The basic formulation is as follows: given a dataset $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, a white-box function g, a loss function l, and constants $\lambda_z > 0$ and $\lambda_{lasso} \geq 0$, find the parameters for $g_1, \cdots g_n$ and embedding $Z \in R^{n \times d}$ that

Table 1
Distribution intervals of the selected variables.

	Variables	Unit	Range		Mean	Description
Chemical composition	C	wt%	0.022	0.204	0.073	
	Mn	wt%	0.086	1.5	0.157	
	P	wt%	0.007	0.09	0.027	
	S	wt%	0	0.018	0.003	
	Si	wt%	0.002	0.27	0.045	
	Nb	wt%	0	0.045	0.004	
	V	wt%	0	0.046	0.008	
	Al	wt%	0	0.03	0.002	
	Cu	wt%	0	0.4	0.038	
Material parameters	FDH	mm	1.4	11.8	3.683	Exit thickness
	RDT	℃	973	1203	1120	Initial rolling temperature
Process parameters	FDT	℃	796	973	891	Finishing rolling temperature
	RT-F1	℃	943	1098	1051	Rolling temperature of F1
	RDR-F4	%	16.1	42.3	32.6	Rack drop rate of F4
	RDR-F5	%	18.2	38.2	29.9	Rack drop rate of F5
	RDR-F6	%	16.2	29.8	25.0	Rack drop rate
	CR	℃/s	5.34	65.4	28.9	Cooling Rate
Physical properties	YS	MPa	194	504	307	Yield Strength
	TS	MPa	251	626	394	Tensile Strength
	ER	%	20.5	54.5	38	Elongation

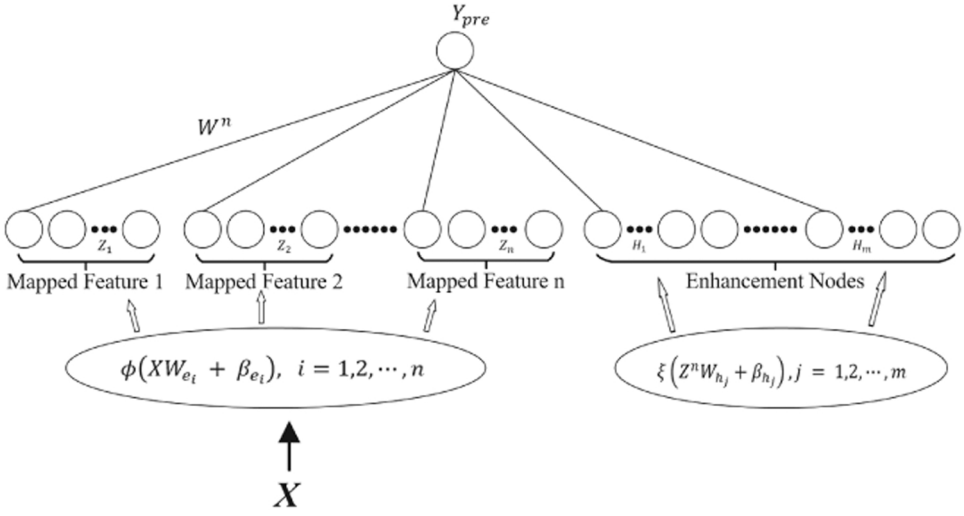


Fig. 5. Architecture of BLS.

minimize the loss given by

$$\ell = \sum_{i=1}^{n}\sum_{j=1}^{m} W_{ij}L_{ij} + \lambda_Z \sum_{i=1}^{n}\sum_{k=1}^{d} Z^2{}_{ik} + \lambda_{lasso}\sum_{i=1}^{n}\sum_{p=1}^{m}|B_{ip}| \tag{1}$$

where $L_{ij} = l(g_i(x_j),\ y_j)$, and $W_{ij} = e^{-D_{ij}}/\sum_{k=1}^{n} e^{-D_{ik}}$, with $D_{ij} = \left(\sum_{k=1}^{d}(Z_{ik} - Z_{jk})^2\right)^{1/2}$. Matrix B is formed by the parameter vectors from all the local models.

Fig. 4 shows the experimental results with SLISEMAP. On the left is the two-dimensional embedding plane, in which the nearby data points can be explained by the same interpretable model. Different colors represent the fidelity of the local models. On the right is a matrix reflecting the coefficients of the local models, in which each row represents a local model and each column is the dataset attribute, and different colors display the coefficients between the attributes and the local models. We can identify that the attributes playing the most significant role in predicting Yield Strength (YS) are the chemical compositions of C and Si, the process parameters of Cooling Rate and Rolling temperature of F1. The dimension of the predicting model is reduced from 39 to 17.

The distribution intervals of the selected 17-dimensional input variables and 3-dimensional output variables for modeling are list in Table 1.

4. Proposed method for mechanical performance prediction of hot-rolling steel

4.1. Broad learning system

As the dimensions and scale of data became large, deep-structure neural networks began to be widely applied in many fields. However, large numbers of weighting and structural parameters of deep-structure neural networks need iterative training process, which is time-consuming and resource intensive. Flat networks such as random vector function linked neural networks (RVFL) overcome these shortcomings, to some extent, by adding nodes laterally and generating hidden-layer weights randomly. The recently emerging broad learning system, proposed by Chen, has become one of the most popular flat networks due to its outstanding efficiency and performance. As shown in Fig. 5, BLS first converts raw data into mapped features through randomly generated weights and stores them in feature nodes. The feature nodes are then expanded into enhanced nodes through randomly generated weights. Feature nodes and enhancement nodes can effectively extract features from data, ensuring the effectiveness of the system on data. By embedding classical low-rank algorithms into generalized learning networks, such as singular value decomposition (SVD) and non-negative matrix factorization (NMF), features extracted by other models can be used to simplify the processing of high-dimensional data.

The training process of BLS is described as follows. Assuming that the training data has N arbitrarily different samples (x_i, y_i), where $x_i = [x_{i1}, x_{i2}, \ldots, x_{in}] \in R^n$, $y_i \in R^n$. For feature nodes, the mapping features of the ith group are generated by a linear mapping function:

$$Z_i = \phi(XW_{e_i} + \beta_{e_i}),\quad i = 1, 2, \ldots, n \tag{2}$$

where W_{e_i} and β_{e_i} are weights and bias coefficients with appropriate dimensions randomly generated as the ith group of feature nodes. The number of feature node groups is n, each group has k feature nodes, and $\phi(\cdot)$ represents linear mapping function. All the feature nodes are combined and recorded as Z^n,

$$Z^n = [Z_1,\ Z_2, \ldots, Z_n] \tag{3}$$

$Z^n \in R^{(N*nk)}$ and nk is the number of all feature nodes. All feature node groups are transformed by nonlinear functions to generate enhanced node groups, and the jth group of enhanced nodes is expressed as follows:

$$H_j = \xi\left(Z^nW_{h_j} + \beta_{h_j}\right),\quad j = 1, 2, \ldots, m \tag{4}$$

where W_{h_j} and β_{h_j} are random weight coefficients of the jth group enhanced node with appropriate dimensions. The number of enhanced nodes is m, and $\xi(\cdot)$represents a nonlinear function. All enhancement nodes can be represented as H^m, and $H^m \in R^{(N*m)}$.

$$H^m = [H_1,\ H_2, \ldots, H_m] \tag{5}$$

Combining enhancement nodes and feature nodes as A,

$$A = \left[Z_1, \ldots, Z_n | \xi\left(Z^nW_{h_1} + \beta_{h_1}\right), \ldots, \xi\left(Z^nW_{h_m} + \beta_{h_m}\right)\right] = [Z^n | H^m] \tag{6}$$

where $A \in R^{N*(nk+m)}$ and $nk+m$ is the total number of nodes for feature nodes and enhancement nodes. Then BLS can be expressed as follows:

$$T = A\beta \tag{7}$$

where $T = [t_1, t_2, \ldots, t_N]^T$ is the predicted output of the sample, and β is the output weight of the BLS, connecting the feature and enhancement nodes to the output layer. β can be obtained by minimizing the ridge regression Eq. (8):

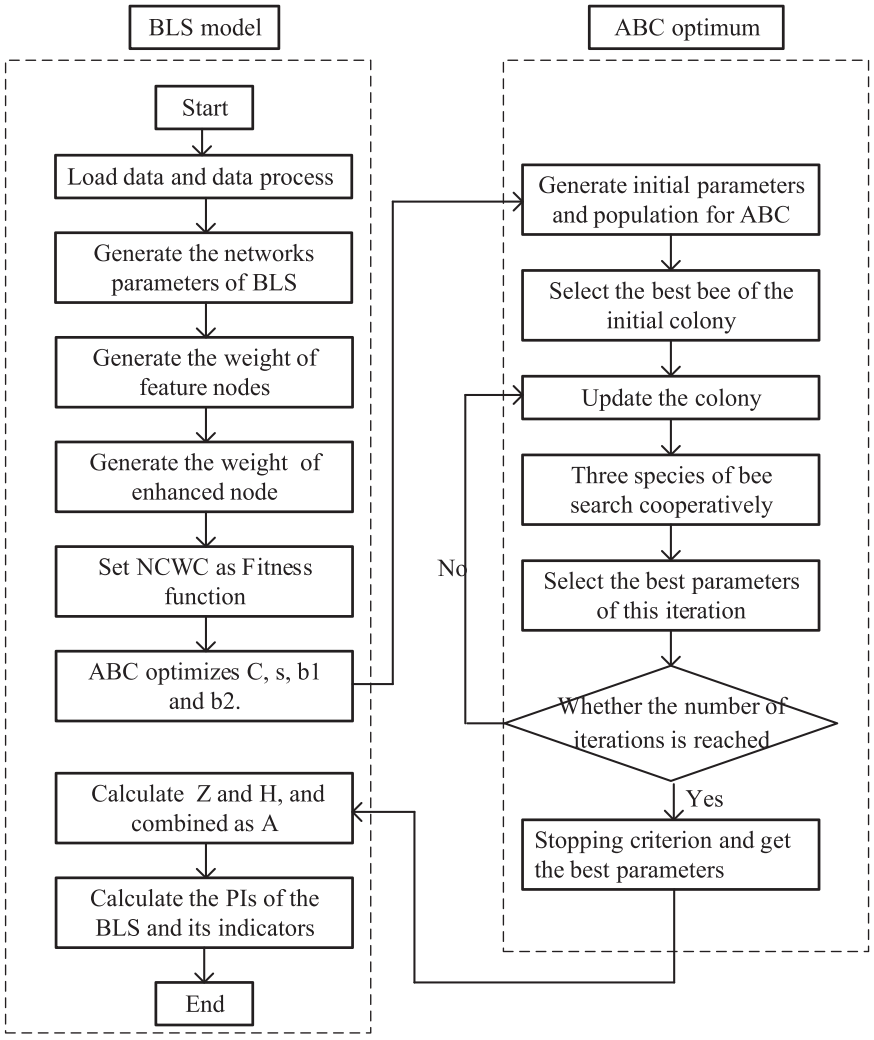


Fig. 6. Overall algorithm flow chart of ABC-BLS.

$$\underset{\beta \in Rn_h * n_o}{\arg min} L_{BLS}\frac{C}{2}\| \beta \|^2 + \frac{1}{2}\| A\beta - Y \|^2 \tag{8}$$

where $n_h = nk + m$, represents the number of all nodes, n_o represents the number of weight nodes, and C represents further constraints of the sum of β. In Eq. (8), the first term is the regularization term for over-fitting, and the second term is the error vector for the ith training sample. Eq. (9) is set to zero with respect to the gradient:

$$\nabla L_{BLS} = \theta\beta + A^T(A\beta - Y) = 0 \tag{9}$$

By transforming the terms, the solution of β can be expressed as follows:

$$\beta = (A^T \quad A + \theta I_{n_h})^{-1} A^T Y \tag{10}$$

where I_{nh} is the identity matrix of a specific dimension.

4.2. LUBE based on BLS

4.2.1. Formulation of PIs

The LUBE method constructs PIs by a neural network with dual output neurons, which is originally proposed for regression tasks. For a neural network, the number of input neurons, hidden layer neurons and output neurons are n, l, and 2, respectively. The output of the ith sample is represented by $o_i = [o_{i1}, o_{i2}]$, and its label value is noted as y_i. The upper and lower bounds of LUBE can be obtained directly by comparing the two outputs, such as Eqs. (11) and (12).

$$U(x_i) = \max\{o_{i1}, o_{i2}\}, \quad i = 1, 2, \ldots, N \tag{11}$$

$$L(x_i) = \min\{o_{i1}, o_{i2}\}, \quad i = 1, 2, \ldots, N \tag{12}$$

where N is the number of samples, the prediction interval of the ith sample can be expressed as $[L(x_i), U(x_i)]$, and the sample label value is expected to lie within the constructed interval with a specified probability, which is described as the PIs nominal confidence (PINC, PINC $= (1-\alpha)\%$), and the value of α denotes the probability of error.

$$P(L(x_i) \le y_i \le U(x_i)) = (1-\alpha)\% \tag{13}$$

In this study, BLS model was used to construct a PI of quality performance based on the LUBE framework. The evaluation indicators of PIs mainly include PI coverage probability (PICP), PI normalized root-mean-square width (PINRW), and a new comprehensive indicator NCWC is introduced.

4.2.2. PIs evaluation indices

4.2.2.1. PICP. The most fundamental feature of PIs is PICP, which represents the probability that the target value will fall within the constructed PI. The definition of PICP is:

$$PICP = \frac{1}{n}\sum_{i=1}^{n} c_i \tag{14}$$

where, c_i is used to record whether the predicted value of the ith sample falls within the corresponding prediction interval (if it falls within the interval, c_i=1; otherwise, c_i=0).

4.2.2.2. PINAW. Another important feature of the PIs is the normalized average width (PINAW), which describes the PI width. Its definition is as below.

$$PINAW = \frac{1}{NR}\sum_{i=1}^{n}(U_i - L_i) \tag{15}$$

where U_i and L_i denote the upper and lower bounds of the constructed PI for the *i*th sample, respectively, and R represents the range of the target values.

4.2.2.3. Comprehensive measure of PI-NCWC. PICP and PINAW are two contradictory evaluation indicators in that a higher PICP usually leads to a wider width, and a narrower width usually leads to a lower PICP. However, an overly broad PI is meaningless to decision-makers. There is a center bias between PIs and label values, which also affects the quality of PIs. Combined with previous work, a new index is proposed to comprehensively consider coverage, width and center deviation, and it is used as an objective function to optimize the width and sharpness of PIs.

$$MPICD = \frac{1}{RN}\sum_{i=1}^{N}\left|\frac{U_i + L_i}{2} - y_i\right| \tag{16}$$

When considering the effect of interval width on the quality of PIs, we can measure the estimation error of PI by introducing Winkler points to calculate the average width, and the width of PIs can be expressed as:

$$G(x_i) = U(x_i) - L(x_i) \tag{17}$$

The Winkler points are defined as follows.

$$S(x_i) = \left\{ \begin{array}{c} -2\alpha G(x_i) - 4(L(x_i) - y_i), \quad if \quad y_i \le L(x_i) \\ -2\alpha G(x_i), \quad if \quad L(x_i) \le y_i \le U(x_i) \\ -2\alpha G(x_i) - 4(y_i - U(x_i)), \quad if \quad y_i > U(x_i) \end{array} \right\} \tag{18}$$

where α is related to the confidence level (α = 0.1 for 90% nominal confidence level). The overall score can be evaluated as:

$$PIES = \frac{1}{N}\sum_{i=1}^{N} S(x_i) \tag{19}$$

The new indicator NCWC can be expressed as:

$$NCWC = |PIES| + (\lambda + MPICD) * e^{\eta * \lambda * (PINC - PICP)} \tag{20}$$

where λ is defined as:

$$\lambda = \left\{ \begin{array}{ll} 0, & PICP \ge PINC \\ 1, & PICP < PINC \end{array} \right\} \tag{21}$$

where η is a hyperparameter that controls the width of PI, which amplifies the difference between PICP and *PINC*, to punish the non-ideal PIs. As discussed above, when PICP is greater than or equal to PINC, the value of NCWC is equal to the sum of the absolute value of PIES plus MPICD; otherwise, the value of NCWC is equal to the sum of the absolute value of PIES and $(1 + MPICD)e^{\eta * \lambda * (PINC - PICP)}$. To have a good NCWC value, the optimization model must obtain relatively large coverage, relatively narrow PIs with small central deviations.

Table 2
Comparison of different algorithms.

Algorithm		PICP	PINAW	NCWC	Time (s)
Delta	BPNN	0.89	0.45	34.30	463.2
	ELM	0.87	0.44	43.88	72.6
LUBE	ABC-BPNN	0.88	0.42	36.18	144.7
	ABC-ELM	0.91	0.37	32.17	75.2
	ABC-SVR	0.90	0.35	27.64	93.4
	ABC-BLS	0.93	0.36	23.36	36.5

4.3. LUBE optimized by ABC

The BLS-based LUBE model is transformed into an optimization problem, as shown below.

$$Minimize : NCWC$$

$$s.t. \quad 0 \le PICP \le 100\% \& PINAW > 0$$

As a kind of swarm intelligence algorithm, the ABC algorithm employs an optimization mechanism inspired by the honey-collecting behavior of a bee colony to realize the sharing and exchange of information. The algorithm was employed to obtain a more accurate prediction value for PIs in this study. Fig. 6 presents the flowchart of the overall algorithm.

Step 1: Load the dataset and preprocess the data. Industrial data often contain noise, and missing values, which lead to serious negative impact on the accuracy and reliability of the model. Therefore, it is necessary to eliminate noise and missing values to improve the robustness of the model.

Step 2: Initialize the parameters of the ABC-BLS model, including the network structure of BLS and parameters of the ABC algorithm. The initialization parameters of ABC include the population size, number of bees, maximum number of iterations, and upper and lower limits of the solution. The fitness function of ABC was constructed based on the comprehensive index NCWC.

Step 3: Obtain the optimal parameters (regular term *C*, enhance node shrinkage ratio *s* and the two biases b_1, b_2 that determine the PIs) by applying the ABC algorithm.

Step 4: Calculate the output of the model to form the corresponding PIs based on (7), (11) and (12). Calculate the indices to evaluate the quality of the PIs of the mechanical performance to evaluate the quality of the ABC-BLS model.

5. Experiments and analysis

The big data platform developed in this study is implemented in a large iron and steel enterprise. The platform covers three continuous casting lines, two heating furnaces, one hot continuous rolling line, one continuous annealing line and one recoiling line. We used 1335 hot-rolling steel coil samples collected through the platform as the dataset for training and testing, of which 85% were used as the training set and 15% (200 coils) were used as the test set. We selected yield strength, one of the most representative mechanical properties, as the predicted value. To test the performance of the proposed method, we implemented two delta-based approaches, namely back propagation neural network (BPNN), extreme learning machine (ELM) and other three LUBE-based approached, namely ABC-BPNN, ABC-ELM, ABC- support vector regression (SVR) as comparative methods. The above methods were compared and analyzed to verify the effectiveness of the proposed LUBE method, ABC-BLS. For a fair comparison, all the parameters involved in the tested LUBE methods were optimized by the ABC algorithm. The parameters of the ABC were as follows: maximum number of iterations to terminate was 2000, population size was 50, the value of limit to control jumping out of the local optimal was 10. The network structure of BPNN and ELM are set to 17–8–3. The number of feature node groups, feature nodes in each group and enhanced nodes of BLS were set to

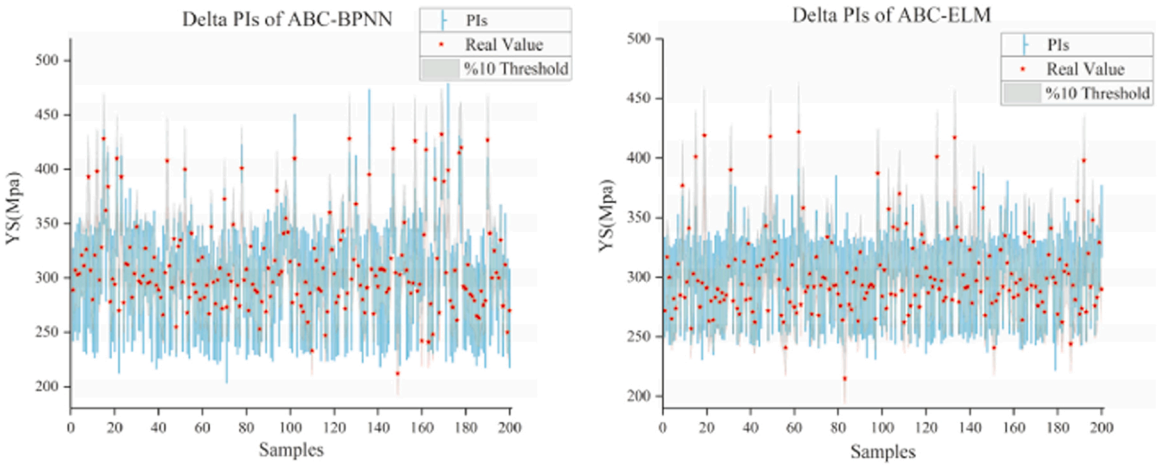


Fig. 7. PIs of Delta methods under 95% confidence level for BPNN and ELM.

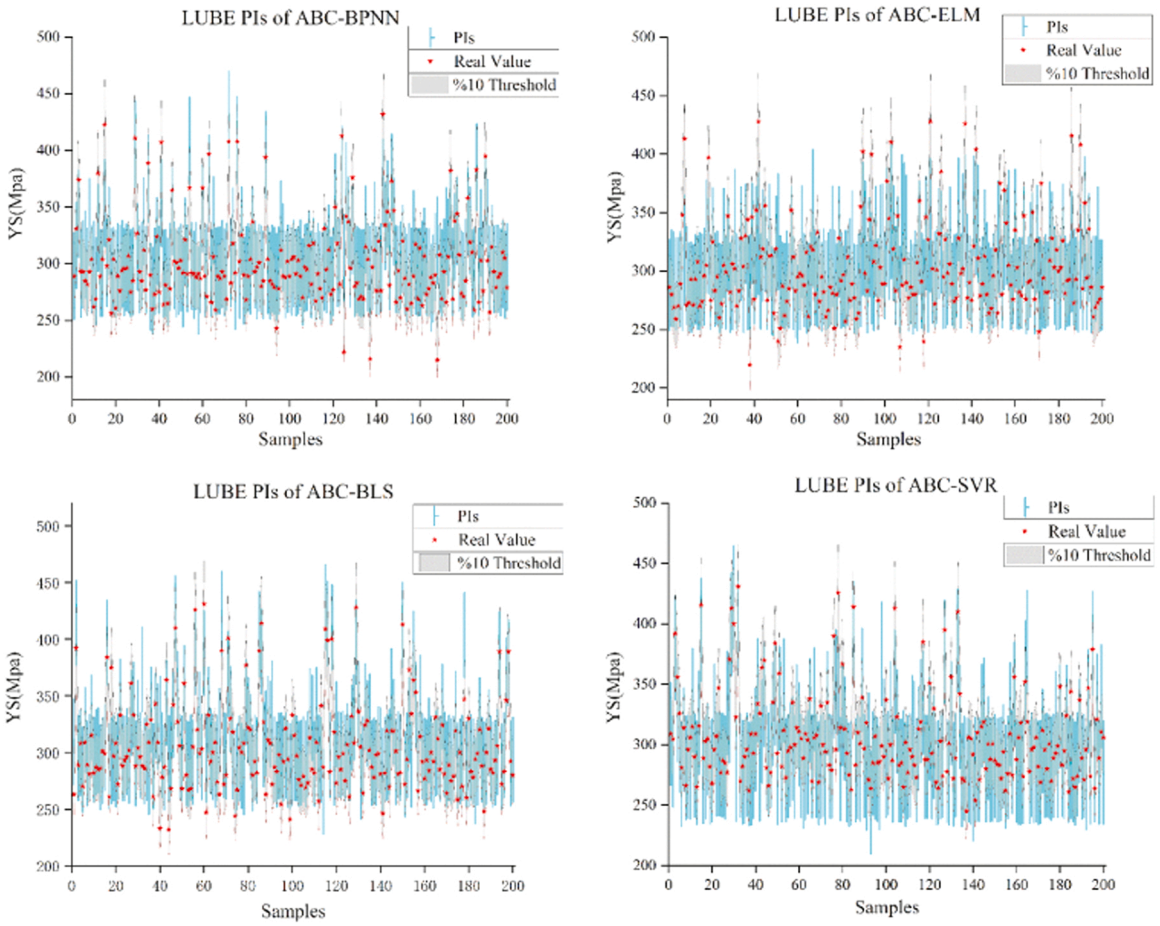


Fig. 8. PIs of LUBE methods.

10–12–50, respectively. During the experiments, the average PI evaluation indices values were calculated through 20 repeated experiments, as shown in Table 2.

The experimental results show that the overall performances of LUBE methods are better than delta methods in PIs quality and running time. PIs constructed by the BPNN and ELM algorithm approach 89% and 87% of the true yield strength value, respectively, which proves that these two approaches perform well in PI coverage rate. However, the width of the prediction intervals obtained by delta methods is large, which reduce the information and the reliability of prediction results. Fig. 7 shows the yield strength PIs at 95% confidence level for the BPNN and ELM, which reflect intuitively the prediction interval. Notably the running time of the ELM algorithm is more than 80% less than that of the BPNN, which proves that the ELM algorithm has a much faster learning speed.

Fig. 8 shows the PIs for the four LUBE-based approaches. The coverage rate of ABC-BPNN, ABC-ELM, ABC-SVR and ABC-BLS is 0.88, 0.91, 0.90, and 0.93, respectively. The PI coverage rate of ABC-ELM,

ABC-SVR and ABC-BLS all exceed 90%, and the interval widths of these three algorithms are less than 0.40. Of the four LUBE methods, ABC-BLS has the highest PI coverage rate, 6.9% larger than ELM in Delta, and 3.3% larger than ABC-SVR in LUBE. In terms of the interval width, ABC-SVR has the narrowest PI, 22.2% smaller than that of BPNN in Delta. NCWC is calculated to comprehensively consider the coverage, width and center deviation of PIs. From the results ABC-BLS has the best performance comprehensively, 46.8% less than ELM in Delta, 27.4% less than ABC-ELM in LUBE, and 15.5% less than ABC-SVR in LUBE. Although ABC-ELM in LUBE avoids the problem of numerical calculation of Jacobian matrix and Hessian matrix, it takes more running time than ELM in delta due to the optimization of network parameters. ABC-BLS takes the least running time since only the weights from the hidden nodes to the output nodes need to be trained. By comprehensive analysis, we can conclude that the proposed ABC-BLS in LUBE is effective and efficient in constructing the PIs.

6. Conclusion

Industrial big data technology is an important driver of intelligent manufacturing. Quality prediction of steel products such as mechanical properties plays a vital role in the steel industry. In this study, the characteristics of steel production data are analyzed, including multi-source, heterogeneity, spatial-temporal hereditary, multi-modal and time-varying. Based on the main features, a four-layer big data platform for steel production is developed. A data-driven approach for constructing the PIs of mechanical performances for hot-rolling strips is proposed to represent the uncertainty and reliability of the prediction results. A new manifold visualization method, SLISEMAP, which combines manifold visualizations and local explanations, is applied to reduce feature dimensions with interpretability. The LUBE method is adopted to construct PIs, in which the recently emerging BLS is used as the basic training network model and the ABC algorithm is applied to optimize the weighting parameters of BLS. A hot-rolling steel coil dataset consisting of 39 variables and 1335 coil samples from a large iron and steel enterprise is used to validate the proposed method, and two delta-based approaches, namely BPNN, ELM and other three LUBE-based approached, namely ABC-BPNN, ABC-ELM, ABC-SVR are implemented as comparison methods. Results show that the proposed ABC-BLS in LUBE is effective and efficient in constructing PIs with a higher coverage probability and a narrower interval width.

In future, we will focus on the online prediction of mechanical properties under different PIs construction framework, such as delta, bootstrap and LUBE. Data processing methods for high-dimensional data, such as outlier detection, feature extraction and clustering will also be investigated. Moreover, process parameter optimization methods based on the prediction results will also be studied to improve the quality indicators in steel production.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. List of abbreviations

Abbreviation	Description
PIs	Prediction Intervals
LUBE	Lower Upper Bound Estimation
BLS	Broad Learning System
ABC	Artificial Bee Colony
BPNN	Back Propagation Neural Network
ELM	Extreme Learning Machine
SVR	Support Vector Regression
MVE	Mean Variance Estimation
PLC	Programmer Logic Controller
DCS	Distributed Control System
MES	Manufacturing Execution System
ERP	Enterprise Resource Planning
OPC	Object Linking and Embedding for Process Control
ODBC	Open Database Connectivity
SQL	Structured Query Language
YS	Yield Strength
TS	Tensile Strength
ER	Elongation
NMF	Non-negative Matrix Factorization
SVD	Singular Value Decomposition
RVFL	Function Linked Neural Networks
PINRW	Prediction Interval Normalized Root-mean-square Width
PINAW	Prediction Interval Normalized Average
PICP	Prediction Interval Coverage Probability
NCWC	New Coverage Width-based Criterion
MPICD	Mean Prediction Interval Center Deviation
PIES	Prediction Interval Evaluation Score

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